

Question	Marking Guidance	Mark	Comments
05.1	Mark as pairs: 1 and 2 OR 3 and 4 1. Deletion/translocation; 2. Could mean triplet(s)/codon(s) missing OR Could mean amino acid(s) missing (from the polypeptide/SURF1); 3. Substitution/inversion/addition/duplication/deletion/translocation; 4. Could result in a (premature) stop triplet/codon;	2 max (2 x AO2)	2. Reject could mean an amino acid is not produced

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05.2	Correct answer of 3 (people) = 3 marks;;; 3.462564706/3.48/3.45 (or any correct rounding down to 1dp) = 2 marks (answer not to the nearest whole number) 23 = 2 marks (number of Faroe Islanders with nuclear mutations) 4 = 2 marks (not factored in that only 80% of mutations are in nuclear DNA) 29 = 1 mark (number of Faroe Islanders with LS)	3 (3 x AO2)	

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05.3	Mark as pairs: 1 and 2 OR 3 and 4 1. Genetic drift; 2. Frequency is higher by chance OR High frequency is not due to natural selection; 3. (Only) inbreeding/interbreeding (within a population) OR No (inter)breeding with other populations OR (Inherited from) common ancestor; 4. Low genetic diversity OR Small gene pool OR Little gene flow OR Higher chance of inheriting <u>allele</u> OR Frequency of <u>allele</u> higher (in offspring);	2 max (2 x AO2)	3. Accept descriptions of inbreeding OR no interbreeding 3. Accept reproductively isolated 3. Accept genetic bottleneck OR founder effect

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05.4	2 max for mark point 1 to 4 OR 5 to 7 Yes (no mark) - Some people could be heterozygous/carriers; - Could prevent (human) suffering/death OR Could allow for (informed) decisions about having children; (But only) in families/people with a history of LS OR (only) in families/people in the Faroe Islands (where high frequency/1: 1700); - Cost of screening might be cheaper than cost of treating LS; No (no mark) - It is rare (globally) OR (Only) 1 in 40 000 (globally); - Caused by (too) many genes/one of 75 genes OR Would need (too) many probes/75 probes; (Too) expensive to produce tests/probes (for more than 75 different genes) OR (Too) expensive to screen all;	3 max (3 x AO3)	

